

Mid Semester Examination 2025

Name of the Program: B. Tech. (ME)
Name of the Course: Solar Energy Engineering
Time: 1.5 Hour

Semester: V
Course Code: TME511
Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks.

Q1	(10 marks)	CO1
(a)	What do you mean by direct radiation, diffused radiation and global radiation. Which type of radiation is used by flat plate collector and concentrator collector? What is solar constant?	
OR		
(b)	Explain different laws of heat transfer. Also, explain overall heat transfer coefficient for the case involving conduction, convection and radiation.	
Q2	(10 marks)	CO1
(a)	Define latitude, declination, hour angle, zenith angle and solar altitude.	
OR		
(b)	How can solar energy engineering help in achieving sustainable development goals of the United Nations. What is the value of efficiency for solar panel currently available in the market? What do you mean by solar time?	
Q3	(10 marks)	CO2
(a)	Explain different kinds of evacuated collectors. What is the advantage of evacuated collector over flat plate collector.	
OR		
(b)	Explain the working of parabolic trough collector and dish Stirling.	
Q4	(10 marks)	CO2
(a)	Classify concentrator collector. What is the effect of nanofluid on the performance of concentrator collector. Write the challenges of using nanofluid in solar system.	
OR		
(b)	What is sensitivity analysis of collector? Explain with suitable example.	
Q5	(10 marks)	CO2
(a)	Write the working of photovoltaic thermal flat plate collector. How much portion of surface area of collector should be covered with PVT and why?	
OR		
(b)	What do you mean by uniform end-of-year annual cost (UEAC)? Calculate UEAC for the solar collector having present value as Rs. 30,000, salvage value as Rs. 5000, life as 25 years and rate of interest as 9% annually. Assume any missing data.	

Note for the question paper setters: (Assuming two units are covered)

- Question paper should have questions from both the units covering the related COs
- Sub questions (a) and (b) can be further divided if required

